



The same pigments

A new study shows that pigments found in eyes of octopi is also found in their skin.
<http://dgit.in/1Q2D36D>

Little Genius

An 11 year old student recently graduated from college saying, "There was nothing to it": <http://dgit.in/1IQNGKD>

on brushes for commutation. Instead, it replaces them with an Electronic Speed Controller (ESC). The digital timing and stepping control not only increases input precision, but it also allows the motor to achieve ridiculously high power and torque output, which is only limited by the thermal ceiling. However, the single most important factor going for brushless motors is durability. The lack of friction and the ability to dissipate heat better significantly increases the lifespan of a brushless motor. The improved efficiency, lightness, and durability of a brushless motor easily offset the additional weight of an ESC.

Choosing the Right Platform

When it comes to choosing parts for your DIY helicopter project, it's important to remember that while this approach gives a near infinite freedom to customise your helicopter, you're still bound to a modular platform determined by the brand and model.

In other words, it's imperative to decide upon a make/model and order the constituent parts accordingly. Helicopters are distinguished according to their size, which typically ranges anywhere between 120 and 700.

The number represents the radius of the main rotor. Greater the number, the larger the helicopter. Since we are primarily interested in micro RC helicopters, it's better to stick to platforms that are smaller than 200.

There are three main players in the micro RC helicopter market. Walkera and E-flite represent value for money offerings, whereas Align products deliver high performance and quality at a commensurately higher premium. The extra money you pay for Align products pays off in terms of powerful ESCs and brushless motors combined with precision mechanical components that allow these micro helicopters to perform advanced acrobatic manoeuvres out of the box.

Those seeking to cut their teeth on less expensive, but equally capable gear from Walkera and E-flite can build their own helicopters from \$100 to \$300, with another \$50 to \$300 for the transmitter, LiPo battery pack, and a battery recharger.

It would be deceptively easy to fill up the remainder of the issue with the virtues of micro RC helicopter models such as Align T-REX 150, E-flite Blade 130X, or Walkera Genius CP followed by a step-by-step process of assembling each one. Since that isn't feasible, the next best option is to visit a micro RC helicopter specialist e-store such as Wow Hobbies [<http://www.wowhobbies.com>] to choose one according to your needs and budget.

Once you have zeroed in on a make/model of choice, you can obtain user manuals, exploded-view blueprints, and other associated schematics from RTF-Heli [<http://www.rtf-heli.com/Downloads.html>]. These documents will also give you a complete idea of the parts needed to build a helicopter from scratch. It must be noted that you're better off buying the same parts from China-based sellers on Ebay (international), most of whom ship to India for free. Unless, of course, you have a benevolent uncle who would agree to haul helicopter parts for you.

Putting It All Together

Now that we have the helicopter parts and schematics at hand, all you need is a set of watchmaker's screwdrivers (you'll mostly need Phillips PH000, PH00, or PH0 sizes), soldering kit (iron, solder, and flux), modelling or thin-nosed pliers, heat-shrink tubing, and double-sided tape. The adjoining images serve as a step-by-step assembly guide for a Walkera 4#3B micro RC helicopter. This is a fixed pitch helicopter, so it has a more complicated rotor head with an additional flybar assembly. Considering the rapid advances in micro RC helicopters and subsequent reduction in prices, chances are that you're going to choose a more sophisticated collective pitch helicopter. This will add a servo, but the rotor head assembly will be a lot less complicated.

At any rate, the assembly instructions imparted here apply to all types of micro RC helicopters being considered by this guide. The mild variations will be restricted to the position of basic parts and polarity of some electrical connections. However, that shouldn't be an issue since you already have access to requisite blueprints for your particular helicopter model.



Read the manual for the rotor head assembly

(1) Depicting the rotor head assembly would be moot, because it varies with each model and manufacturer. You can't really mix and match parts for a rotor head, so you're essentially going to buy it as a kit. The idea is to assemble it according to the instructions provided with the package or the same depicted in the schematics. The rotor head assembly itself is attached to the primary shaft using a small pin. The shaft then eases snugly into the helicopter frame through a pair of bearings. The position of the bearings and shaft will vary according to your helicopter model, but it should roughly resemble what's depicted in the image.



Main rotor gear assembly



Attaching the rotor gear with a pin

(2) Attach the main rotor gear to the part of the primary shaft protruding from the bottom of the frame. Now secure the shaft into the gear using a retention pin as shown in the image.

(3) Now to put together the tail rotor assembly and attach it to the frame. Install the tail motor in the tail rotor housing, screw the rudder onto the